

Task System Backend (NestJS) - 30 Day Plan

Goal: Build a production-like backend system using NestJS with multi-tenant architecture, authentication, caching, async processing, real-time updates, file export, and AI integration.

Daily Plan:

Day 1: Setup NestJS project, folder structure, ConfigModule with validation, Prisma init

Day 2: Design database schema (User, Workspace, Membership, Task, ActivityLog)

Day 3: Implement Prisma models + migrations + seed data

Day 4: User module (basic CRUD + password hashing)

Day 5: Auth (register/login + JWT access token)

Day 6: Refresh token + logout flow

Day 7: Auth guards + cleanup

Day 8: Workspace module (create/join)

Day 9: Membership + roles (OWNER/MEMBER)

Day 10: RBAC guard implementation

Day 11: Task CRUD with workspace isolation

Day 12: Task filtering + pagination

Day 13: Query optimization

Day 14: Refactor API + error handling

Day 15: Activity log implementation

Day 16: Logging task changes

Day 17: DB transactions

Day 18: Redis setup

Day 19: Cache task list

Day 20: Cache invalidation

Day 21: Cache consistency testing

Day 22: Queue system setup (BullMQ)

Day 23: Export tasks job (CSV)

Day 24: File download API

Day 25: WebSocket setup

Day 26: Real-time updates

Day 27: Optimize event flow

Day 28: Logging + error classification

Day 29: AI integration (summarization/query)

Day 30: Final refactor + README system design documentation